KARLOSH YADAV

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Education

Indian Institute of Science, Bangalore 7.05/10
Master of Technology, Computer Science and Engineering *Aug. 2025 – June 2027*

Visvesvaraya Technological University 8.75/10
Bachelor of Engineering, Computer Science and Engineering *Aug. 2020 – June 2024*

Internship Experience

Google IT Services India Private Limited – Software Engineering Intern May 2026 – Jul 2026

- Modernized a critical datacenter validation framework by migrating its hard-coded complex structure to a configuration-driven generic architecture using dynamic & reflection-based mechanisms, implementing consistent validation while maintaining exact parity with legacy requirements.
- Designed and implemented a scalable pipeline, analysed pattern of existing system, reduced engineering overhead, implemented modular design method to achieve better latency and efficiency.

Projects

Benchmarking Deep Learning Models for Brain Tumor Classification: A Comparative Study of YOLOv8, ResNet50, and MobileNetV3 Aug 2025 – Dec 2025

- Conducted a comparative analysis of state-of-the-art deep learning architectures for MRI-based brain tumor classification to evaluate performance across model families (YOLOv8, ResNet50, MobileNetV3).
- Performed systematic hyperparameter optimization and ablation studies to analyze factors influencing model accuracy and generalization.
- Applied K-fold cross-validation and rigorous evaluation to ensure statistical reliability and reproducibility.
- Developed a TensorFlow training and evaluation pipeline ensuring stable training and fair model comparison.

Heart Disease Prediction System (Ensemble Learning) Sep 2023 – May 2024

- Built a prediction system using a Soft Voting ensemble of Logistic Regression, Random Forest, and XGBoost.
- Performed comprehensive data preprocessing including missing value handling, categorical encoding, feature scaling, and train-test splitting.
- Addressed class imbalance using SMOTE and optimized models using K-fold cross-validation and GridSearchCV.

Machine Learning Algorithms from Scratch

- Implemented supervised learning algorithms from scratch using NumPy, including Linear Regression, Logistic Regression, Decision Tree, and Random Forest without relying on high-level ML libraries.
- Derived and implemented mathematical formulations such as MSE loss, Binary Cross-Entropy, Gradient Descent optimization, and Information Gain for model training and splitting.
- Applied Random Forest with bootstrap sampling and feature randomization for improved generalization.

Skills

Programming Languages: C++, Java, Python, SQL, HTML, CSS, JavaScript
Skills: Data Structures and Algorithms, Object-Oriented Programming (OOP), DBMS, Software Development, Framework Design, Configuration & Reflection Systems, Diff & Load Testing, System Design (Basics), ML, DL
Languages: English, Hindi, Maithili, Nepali

Certifications

Graduate Record Examination : 333/340
C++ for Everyone: University of California (2021)
Machine Learning for All: University of London (2023)

Extra-Curricular Activities

President: International Student Club (ISC) (04/2023 – 04/2024)
Secretary: Soft Research Computing Society (04/2022 – 04/2023)
Chess: Inter-Institution level player
Cricket: College and District Cricket Team.